Work, Energy and Power — Worked Solutions (Page 2)

Numericals Worksheet, Grade IX (CBSE) · Questions 19–45 · NCERT Chapter: Work and Energy

Formulas used on this page

Idea	Formula	Notes
Work (force at angle θ)	$W = F\, s \cos\theta$	Q19–20 only; extension beyond NCERT IX
Kinetic energy	$E_k = \frac{1}{2} m v^2$	m in kg, v in m/s
Potential energy	$E_p = m g h$	h = height above reference level
Work–energy theorem	$W = \frac{1}{2} m v^2 - \frac{1}{2} m u^2$	Work done = change in K.E.
Conservation of energy	$E_p + E_k$ = constant	Free fall, no air resistance
Speed after falling h	$v = \sqrt{2 g h}$	from $m g h = \frac{1}{2} m v^2$ (m cancels)

Unit conversions to remember: 1 g = 1/1000 kg · 1 cm = 1/100 m · km/h → m/s: multiply by 5/18 (e.g. 72 km/h = 72 × 5/18 = 20 m/s). Take **g = 10 m/s²** unless stated otherwise.

Solutions

Q19. A suitcase is pulled along a floor by a force of 80 N acting at 30° to the horizontal for 15 m. Find (a) the horizontal component of the force and (b) the work done. (cos 30° = 0.866)

Given: F = 80 N, θ = 30°, s = 15 m.

(a) Horizontal component = F cos θ = 80 × 0.866 = 69.28 N

(b) Only the horizontal component moves the suitcase: W = (F cos θ) × s = 69.28 × 15

Answer: (a) 69.28 N (b) 1039.2 J

Note: The vertical component (F sin 30°) is perpendicular to motion, so it does no work.

Q20. A 200 N force acts on a cart at 60° to its direction of motion and does 5000 J of work. Find the displacement.

Given: F = 200 N, θ = 60° (cos 60° = 0.5), W = 5000 J.

W = F s cos θ ⇒ s = W ÷ (F cos θ) = 5000 ÷ (200 × 0.5) = 5000 ÷ 100

Answer: s = 50 m

Q21. Find the kinetic energy of a 2 kg body moving with a velocity of 10 m/s.

Given: m = 2 kg, v = 10 m/s.

$E_k = \frac{1}{2} m v^2 = \frac{1}{2} \times 2 \times 10^2 = \frac{1}{2} \times 2 \times 100$

Answer: Ek = 100 J

Q22. A cricket ball of mass 0.16 kg is bowled at 25 m/s. Calculate its kinetic energy.

Given: m = 0.16 kg, v = 25 m/s.

$E_k = \frac{1}{2} \times 0.16 \times 25^2 = 0.08 \times 625$

Answer: Ek = 50 J

Q23. A car of mass 1000 kg is moving at 72 km/h. Find its kinetic energy.

Given: m = 1000 kg, v = 72 km/h = 72 × 5/18 = 20 m/s (convert first!).

Ek = ½ × 1000 × 20^2 = 500 × 400

Answer: Ek = 200,000 J = 2 × 10^5 J

Q24. A body of mass 8 kg has a kinetic energy of 400 J. Find its speed.

Given: m = 8 kg, Ek = 400 J.

Ek = ½ m v^2 ⇒ 400 = ½ × 8 × v^2 = 4 v^2

v^2 = 400 ÷ 4 = 100 ⇒ v = √100

Answer: v = 10 m/s

Q25. The speed of a 5 kg body is increased from 4 m/s to 8 m/s. Find (a) initial K.E. (b) final K.E. (c) how many times the K.E. increases.

(a) Initial Ek = ½ × 5 × 4^2 = ½ × 5 × 16 = 40 J

(b) Final Ek = ½ × 5 × 8^2 = ½ × 5 × 64 = 160 J

(c) Ratio = 160 ÷ 40 = 4

Answer: (a) 40 J (b) 160 J (c) 4 times

Note: Since Ek ∝ v^2, doubling the speed makes the kinetic energy 2^2 = 4 times.

Q26. Body A: 4 kg, 6 m/s. Body B: 12 kg, 2 m/s. Find the ratio of K.E. of A to that of B.

Ek(A) = ½ × 4 × 6^2 = ½ × 4 × 36 = 72 J

Ek(B) = ½ × 12 × 2^2 = ½ × 12 × 4 = 24 J

Ek(A) : Ek(B) = 72 : 24

Answer: 3 : 1

Note: Body B is three times heavier, but A is three times faster — speed matters more because it is squared.

Q27. A bullet of mass 20 g is fired with a speed of 300 m/s. Calculate its kinetic energy.

Given: m = 20 g = 20 ÷ 1000 = 0.02 kg, v = 300 m/s.

Ek = ½ × 0.02 × 300^2 = 0.01 × 90,000

Answer: Ek = 900 J

Q28. Calculate the potential energy of a 10 kg object placed at a height of 5 m above the ground.

Given: m = 10 kg, h = 5 m, g = 10 m/s^2.

Ep = m g h = 10 × 10 × 5

Answer: Ep = 500 J

Q29. A boy of mass 50 kg climbs a staircase of 20 steps, each 20 cm high. Find the gain in his potential energy.

Total height h = 20 × 20 cm = 400 cm = 4 m.

Ep = m g h = 50 × 10 × 4

Answer: Gain in Ep = 2000 J

Q30. An object of mass 20 kg has 1200 J of potential energy. At what height is it above the ground?

Ep = m g h ⇒ h = Ep ÷ (m g) = 1200 ÷ (20 × 10) = 1200 ÷ 200

Answer: h = 6 m

Q31. A water tank holds 2000 kg of water and is placed on a tower 15 m high. Find the potential energy of the water.

Given: m = 2000 kg, h = 15 m.

Ep = m g h = 2000 × 10 × 15

Answer: Ep = 300,000 J = 3 × 10^5 J

Q32. A body raised to a height of 12 m has a potential energy of 480 J. Find its mass.

Ep = m g h ⇒ m = Ep ÷ (g h) = 480 ÷ (10 × 12) = 480 ÷ 120

Answer: m = 4 kg

Q33. A 1.5 kg book is moved from a table 0.8 m high to a shelf 2.3 m high. Find the increase in its potential energy.

Change in height Δh = 2.3 − 0.8 = 1.5 m.

Increase in Ep = m g Δh = 1.5 × 10 × 1.5

Answer: Increase in Ep = 22.5 J

Note: Only the change in height matters, not the height measured from the floor.

Q34. A 10 kg object is lifted 5 m. Compare its potential energy on the Earth (g = 10 m/s^2) with that on the Moon (g = 1.6 m/s^2).

On Earth: Ep = 10 × 10 × 5 = 500 J

On the Moon: Ep = 10 × 1.6 × 5 = 80 J

Ratio Earth : Moon = 500 : 80 = 25 : 4 (Earth value is 6.25 times larger)

Answer: Earth: 500 J; Moon: 80 J

Note: Mass stays the same everywhere; Ep is smaller on the Moon because its g is smaller.

Q35. A force does 250 J of work on a 5 kg body that is initially at rest. Find the final speed.

Work–energy theorem: work done = gain in K.E. Initial K.E. = 0 (at rest).

250 = ½ × 5 × v^2 ⇒ v^2 = 250 × 2 ÷ 5 = 100 ⇒ v = 10

Answer: v = 10 m/s

Q36. The velocity of a 2 kg body changes from 3 m/s to 7 m/s. Calculate the work done on the body.

W = ½ m v^2 − ½ m u^2 = ½ × 2 × (7^2 − 3^2) = 1 × (49 − 9)

Answer: W = 40 J

Note: Common mistake: do NOT use $(7 − 3)^2$ = 16. Square each speed separately.

Q37. A car of mass 800 kg speeds up from 10 m/s to 20 m/s. Find the work done by the engine.

W = ½ m (v^2 − u^2) = ½ × 800 × (20^2 − 10^2) = 400 × (400 − 100) = 400 × 300

Answer: W = 120,000 J (1.2 × 10^5 J)

Q38. A 1500 kg car moving at 20 m/s is brought to rest by brakes in a distance of 50 m. Find the average braking force.

Initial K.E. = ½ × 1500 × 20^2 = 750 × 400 = 300,000 J; final K.E. = 0.

Work done by brakes = loss of K.E. = 300,000 J (F × s = 300,000)

F = 300,000 ÷ 50

Answer: Average braking force = 6000 N

Note: Work done by the brakes is actually −300,000 J (force opposite to motion); the magnitude of the force is 6000 N.

Q39. A bullet of mass 20 g moving at 250 m/s penetrates 10 cm into a wooden block and stops. Find the average resistive force.

m = 0.02 kg, u = 250 m/s, v = 0, s = 10 cm = 0.1 m.

Loss of K.E. = ½ × 0.02 × 250² = 0.01 × 62,500 = 625 J

F × s = 625 ⇒ F = 625 ÷ 0.1

Answer: F = 6250 N

Q40. A constant force of 50 N acts on a 4 kg body at rest on a smooth horizontal surface and moves it 4 m. Find the final speed.

Work done W = F × s = 50 × 4 = 200 J (smooth surface: no friction).

All this work becomes K.E.: 200 = ½ × 4 × v² = 2 v² ⇒ v² = 100

Answer: v = 10 m/s

Q41. A 0.5 kg ball rolling at 12 m/s is stopped by friction after moving 24 m. Find (a) work done by friction (b) frictional force.

Initial K.E. = ½ × 0.5 × 12² = 0.25 × 144 = 36 J; final K.E. = 0.

(a) Work done by friction = change in K.E. = 0 − 36 = −36 J

(b) |W| = F × s ⇒ F = 36 ÷ 24 = 1.5 N

Answer: (a) −36 J (b) 1.5 N

Note: Negative because friction acts opposite to the motion.

Q42. A 2 kg ball is dropped from a height of 20 m. Find its speed just before it hits the ground. (Ignore air resistance)

Conservation of energy: P.E. at top = K.E. at bottom.

m g h = ½ m v² ⇒ v = √(2 g h) = √(2 × 10 × 20) = √400

(Check: Ep = 2 × 10 × 20 = 400 J = ½ × 2 × 20² ✓)

Answer: v = 20 m/s

Note: The mass cancels out — a 2 kg ball and a 20 kg ball reach the ground at the same speed.

Q43. A stone of mass 0.5 kg is dropped from a height of 45 m. Find (a) its K.E. and (b) its speed on reaching the ground.

(a) K.E. at ground = P.E. at top = m g h = 0.5 × 10 × 45 = 225 J

(b) ½ × 0.5 × v² = 225 ⇒ v² = 225 × 4 = 900 ⇒ v = 30 m/s (or v = √(2 × 10 × 45) = √900)

Answer: (a) 225 J (b) 30 m/s

Q44. A 4 kg object is released from rest at a height of 10 m. Find its P.E. and K.E. when it is 4 m above the ground.

Total mechanical energy = P.E. at the top = 4 × 10 × 10 = 400 J (K.E. = 0 at rest).

At 4 m: P.E. = m g h = 4 × 10 × 4 = 160 J

K.E. = Total − P.E. = 400 − 160 = 240 J (check: P.E. + K.E. = 400 J ✓)

Answer: P.E. = 160 J; K.E. = 240 J

Note: The 240 J of K.E. equals the P.E. lost in falling 6 m: 4 × 10 × 6 = 240 J.

Q45. A ball (mass 1 kg, as written on the sheet) is thrown vertically upwards with a speed of 20 m/s. Using energy conservation, find the maximum height.

At launch: K.E. = ½ × 1 × 20² = 200 J; at the highest point speed = 0, so K.E. = 0.

All K.E. becomes P.E.: m g h = 200 ⇒ 1 × 10 × h = 200

Shortcut: h = v² ÷ (2g) = 400 ÷ 20

Answer: Maximum height = 20 m

Note: The mass is not actually needed: ½ m v² = m g h, so m cancels and h = v²/2g for any mass.

Answer summary

Q	Answer	Q	Answer	Q	Answer
19	(a) 69.28 N (b) 1039.2 J	28	500 J	37	1.2×10^5 J
20	50 m	29	2000 J	38	6000 N
21	100 J	30	6 m	39	6250 N
22	50 J	31	3×10^5 J	40	10 m/s
23	2×10^5 J	32	4 kg	41	(a) −36 J (b) 1.5 N
24	10 m/s	33	22.5 J	42	20 m/s
25	40 J, 160 J, 4×	34	500 J ; 80 J	43	(a) 225 J (b) 30 m/s
26	3 : 1	35	10 m/s	44	PE 160 J; KE 240 J
27	900 J	36	40 J	45	20 m

Exam tips (CBSE)

• Convert units first: grams → kg, cm → m, km/h → m/s (× 5/18).
• K.E. depends on v^2: doubling speed → 4 times K.E.; tripling speed → 9 times K.E.
• Work–energy theorem: $W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$. Square each speed separately — never $(v - u)^2$.
• Stopping problems (brakes, bullet, friction): average force = K.E. lost ÷ stopping distance.
• Free fall: P.E. + K.E. stays constant; $v = \sqrt{2gh}$ and $h = v^2/2g$ do not depend on mass.
• All 27 answers above agree with the answers printed on the worksheet.
  • Q46–50 are only partly visible in the photo (on the next sheet), so they are not solved here. Send a clear photo of that page to get them solved too.